

SolaX X1/X3-EVC G1 system matching instructions

— for Connecting with SolaX Inverter Use



1 Overview

The Smart EV Charger can communicate with SolaX inverters to form an intelligent photovoltaic, storage and EV charging energy system, thus maximizing the utilization of photovoltaic energy.

The current first-generation EV Chargers from SolaX Power are available in three different versions: Home Edition, Commercial Edition, and Fusion Edition. This article will provide a detailed introduction to the installation and configuration of these three Editions of EV Chargers in systems working with SolaX Inverters.

2 Key Features

2.1 Intelligent charging modes

Thanks to the communication between the charger and inverter, the system has Green/Eco mode to use full-green/Semi-green energy to charge your vehicle.

2.1.1 Green mode

The main purpose of Green mode is to charge the EV with renewable energy as much as possible.

The default level is 6A, in which the Smart EV Charger will only start to charge if generating current reaches 6A, and it never take electricity from the grid.

While there is another 3A level start to charge if generating current reaches 3A, capable to purchase other up to 3A from the grid. This work mode will spend all its effort to help clients use renewable energy and reduce the cost of buying electricity from the grid.

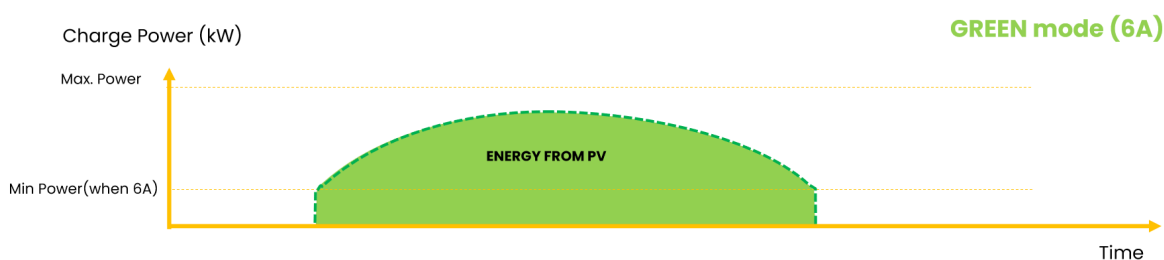


Figure 2 Green Mode 6A

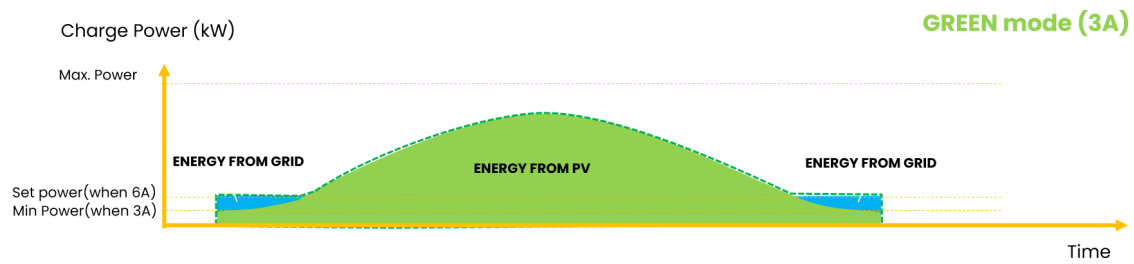


Figure 2 Green Mode 3A

2.2.2 Eco mode

ECO mode help users to charge EV use renewable energy as much as possible while staying above the set minimum efficiency. Electricity from the grid will help maintain the set current.

A charging current can be set thus control the minimum efficiency. For example, the users set the charging current 16A. If the current from the inverter is only 10A then the rest would be taken from the grid as 6A. If the current from the inverter is 18A, then the Smart EV Charger will output 18A.

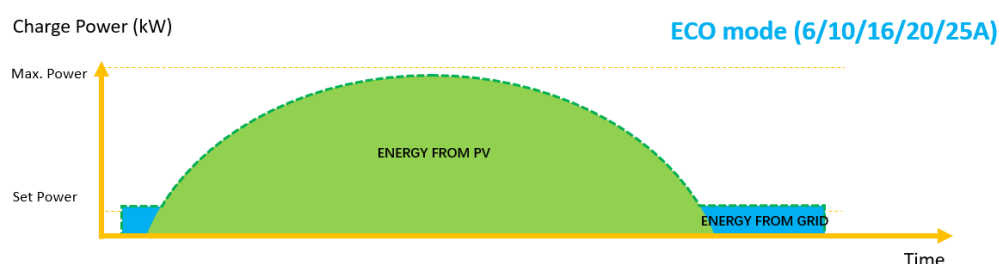


Figure 3 ECO Mode

2.2.3 Fast mode

In Fast mode, the Smart EV Charger will output with the maximum current no matter electricity from the inverter or the grid. This mode usually applies to the users who wants to charge their EV as soon as possible or to the area where there are peak and valley price of utility.

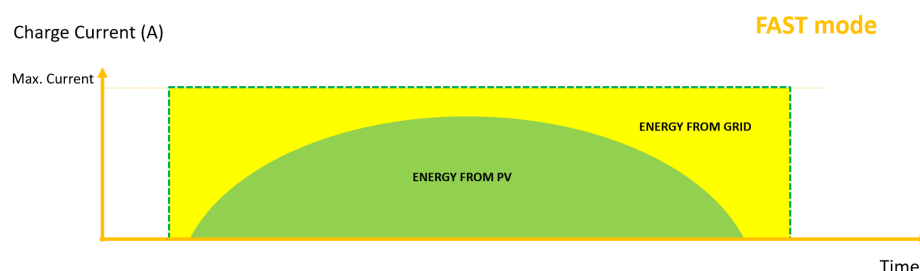


Figure 4 Fast Mode

2.2.4 Timer boost

When users enable the "Timer Boost" function, they can define a specific time period during which the Smart EV Charger will maximize charging speed by utilizing electricity from both the PV system and the grid, regardless of the current work mode. This feature is particularly useful in regions with peak and off-peak electricity pricing, allowing users to schedule charging during low-cost periods to fully charge their EV at minimal expense.

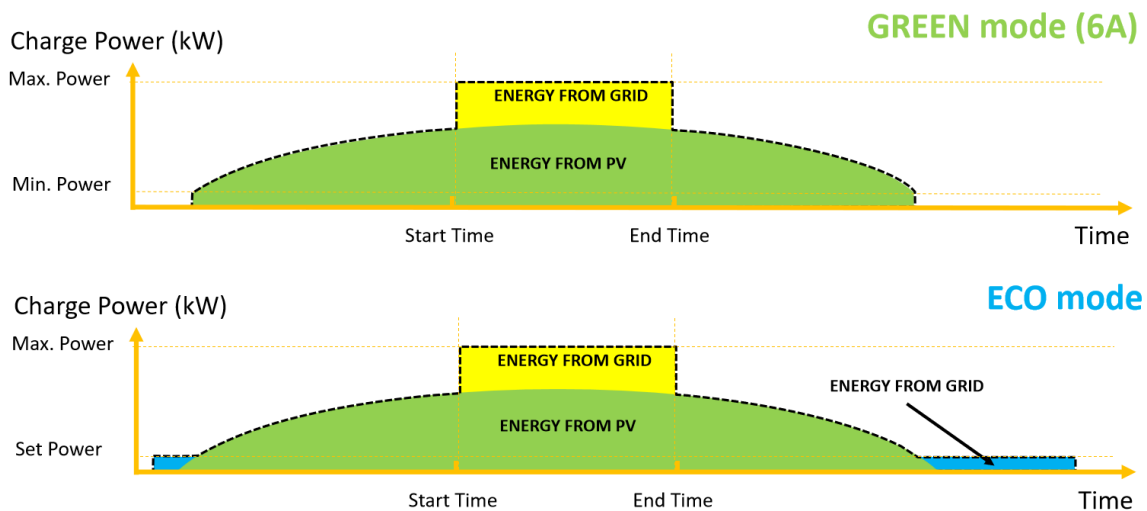


Figure 5 Timer Boost

2.2.5 Smart Boost

With Smart Boost function, the Smart EV Charger will spend all its effort to make sure to finish charging before set time, while using PV energy as much as possible. Users could set an "End Time" and "Charge Energy", the Smart EV Charger will automatically output the power according to the rest time and rest energy.

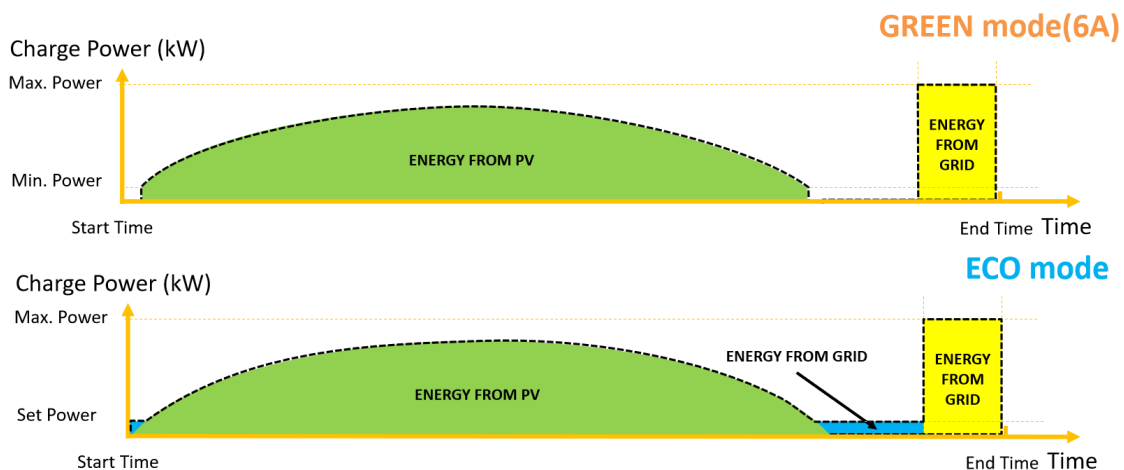


Figure 6 Smart Boost

2.2.6 Schedule charging

User can set schedule charging period to reserve charging session for the electrical vehicle during the off peak time while the electricity price is relatively lower. This can help the user to save the electricity cost and optimize the energy usage. At the same time, the feature could help to reduce the stress on power grid.

This feature can combine with charging power limitation and dynamic load balance function to keep the charging process safe.

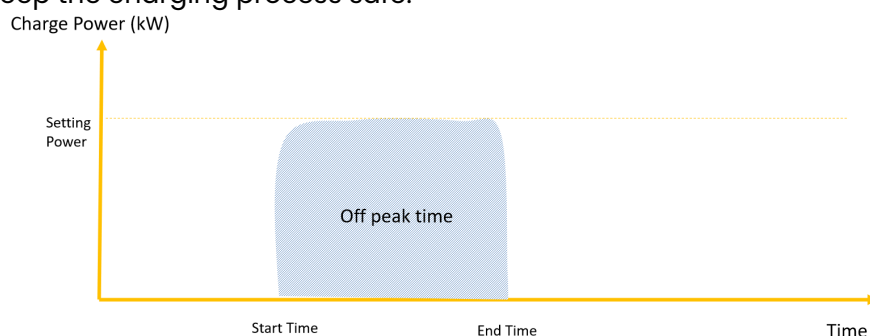


Figure 7 Schedule Charging

2.2.7 Dynamic load balancing function

This function will make sure that the total power of house will not exceed the main grid capacity when using EV charger. It is realized by adjusting the charger power output automatically according to the value of the main grid current.

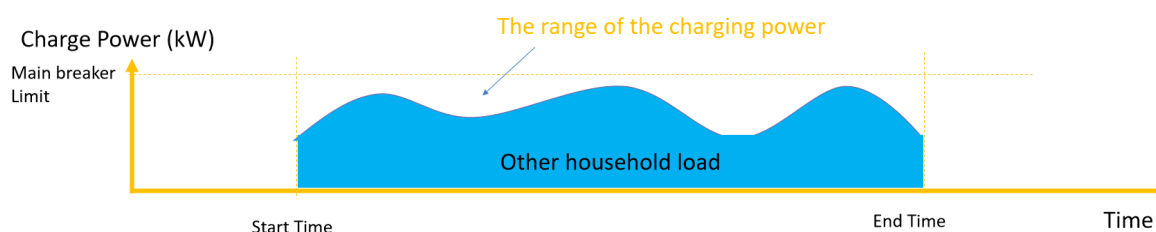


Figure 8 Dynamic Load Balance

3 Communication

3.1 For single EV Charger System

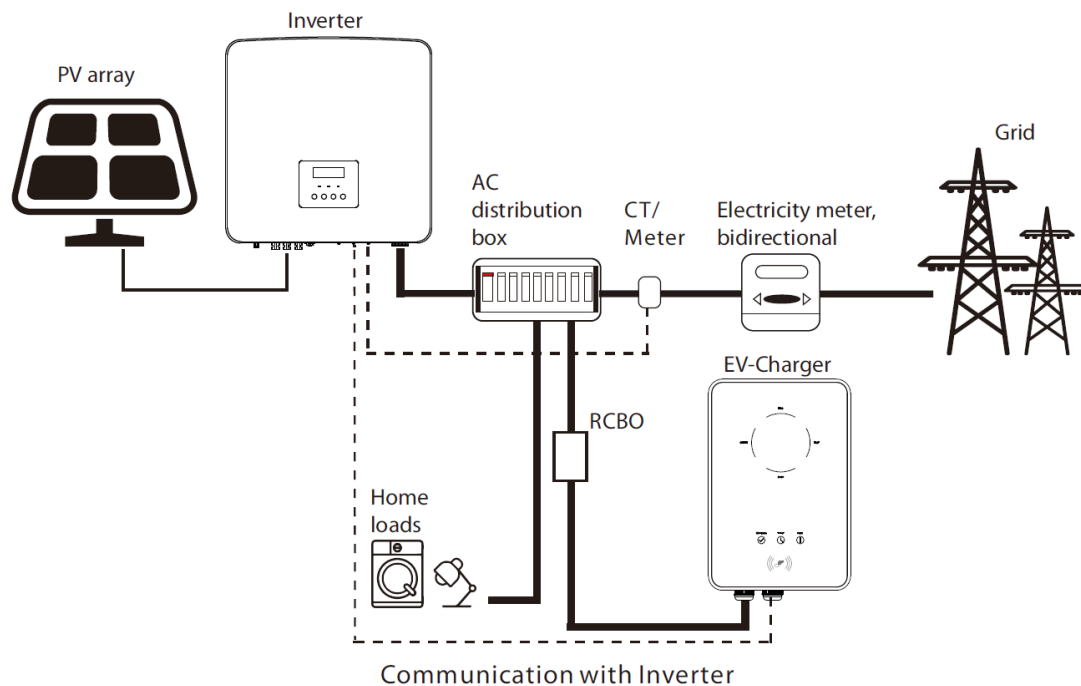


Figure 9 EVC Communicates with SolaX Inverter

In general, the EV Charger is always connected like a normal load in the system. And in such a way that its load can also be recorded via the SolaX Meter/CT, which is connected to inverter.

For the ECO and Green modes or for solar surplus charging, the EV Charger requires information about PV surpluses that are normally fed into the public grid.

Supported Edition:

- Home Edition
- Commercial Edition
- Fusion Edition

3.1.1 Communication Connection for Home Edition and Commercial Edition

The details of the communication connection with SolaX inverter

are as follows.

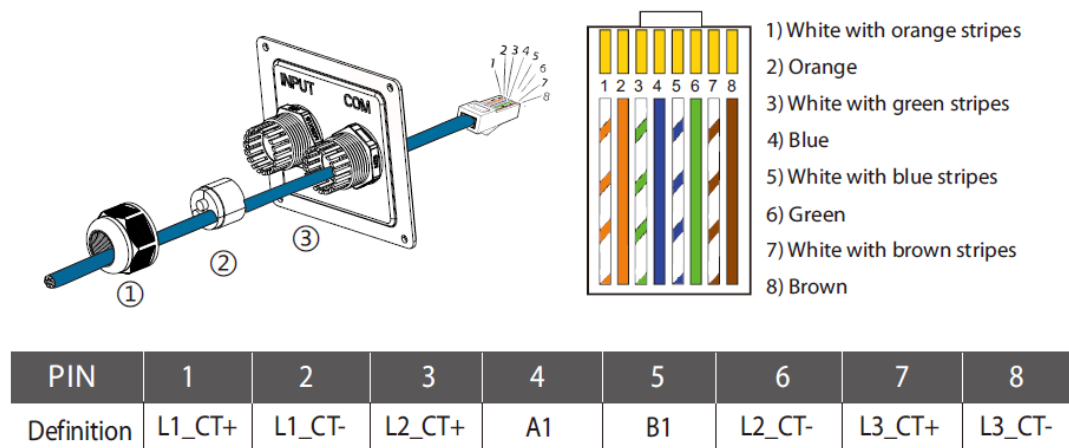


Figure 10 Home and Commercial Edition EVC RJ45 Terminal Pin Definition

- PIN 3, 6, 7, 8 is only for X3-EVC CT solution. Pin4&5 is for connecting inverter or meter.
- It is recommended that customers use a 2-pin network cable to connect the inverter and EV Charger.
- If the inverter is an X3-Hybrid G4, only a 2-pin communication cable can be used for the connection.

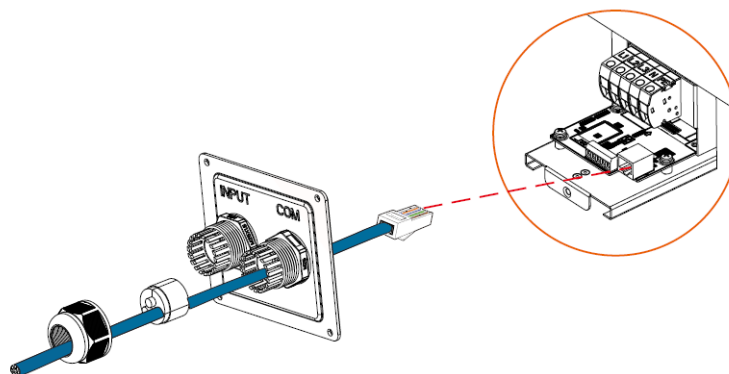
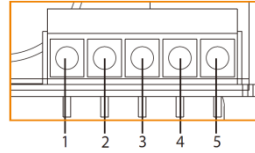


Figure 11 RJ45 Terminal Wiring

- The communication cable can be connected to the RJ45 port as shown in Figure 9, or to the green terminal's Pin 1 (A1) and Pin 2 (B1) as illustrated in Figure 10.
- During installation, ensure that the power cable and communication cable are connected separately to the Input and Com ports, respectively, to prevent the power cable from affecting communication quality.



PIN	1	2	3	4	5
Definition	A1	B1	GND	A2	B2

Figure 12 Home Edition EV Charger RS485 Port

3.1.2 Communication Connection for Fusion Edition

The details of the communication connection with SolaX inverter are as follows.

Pin	1	2	3	4	5	6	7	8
Pin Definition	L1_CT+	L1_CT-	L2_CT+	A1	B1	L2_CT-	L3_CT+	L3_CT-

Figure 13 Fusion Edition RJ45 Terminal Definition

- PIN 3, 6, 7, 8 is only for X3-EVC CT solution. Pin4&5 is for connecting inverter or meter.
- It is recommended that customers use a 2-pin network cable to connect the inverter and EV Charger.
- If the inverter is an X3-Hybrid G4, only a 2-pin communication cable can be used for the connection.
- The communication cable can only be connected to the RJ45 port.
- During installation, ensure that the power cable and communication cable are connected separately to the Input and Com ports, respectively, to prevent the power cable from affecting communication quality.

3.2 For parallel solution

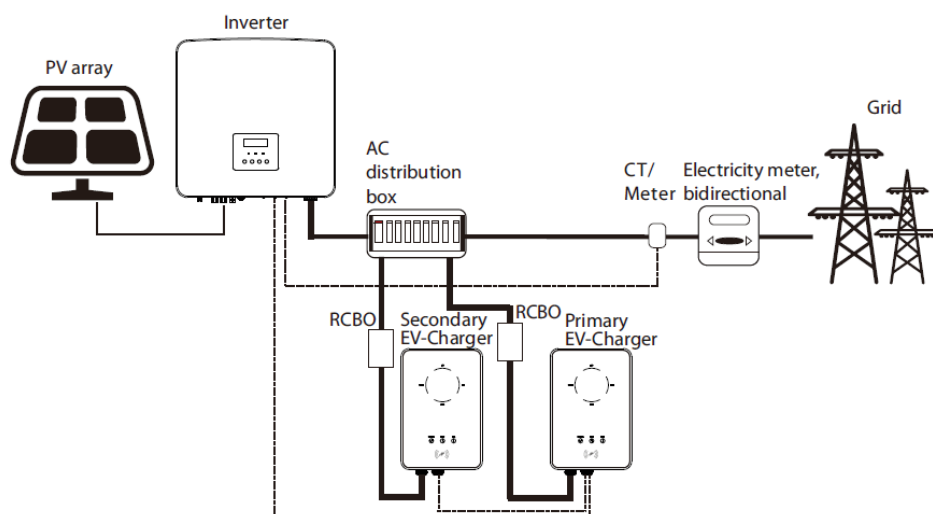


Figure 14 Parallel Solution

For the users who like to install two EV Chargers under same circuit, they can use the parallel function. One EV Charger can be distinguished as primary EV Charger and the other secondary. The primary EV Charger takes the responsibility to collect PV feed in information and grid energy consumption information together and to allocate the PV energy and remaining load capacity for the system according to allocation ratio. The secondary EV Charger's work mode setting will be copied from the primary EV Charger.

Supported Edition :

- Home Edition
- Fusion Edition

3.2.1 Communication Connection for Home Edition

- Use two-core cable to connect the primary EV-Charger's **A1 & B1 of RS485 port** or **RJ45 port** to the inverter's **COM or RS485 port** according to the definitions of the communication ports of the specific inverter.
- Use two-core cable to connect the primary EV-Charger's **A2 & B2** to the secondary EV-Charger's **A1 & B1**.

3.2.2 Communication Connection for Fusion Edition

- Use two-core cable to connect the primary EV-Charger's **A1 & B1 of RJ45 port** to the inverter's **COM or RS485 port** according to the definitions of the communication ports of
- the specific inverter.
- Use two-core cable to connect the primary EV-Charger's **A2 & B2** to the secondary EV-Charger's **A1 & B1**.

3.3 Ports used by different inverters for communication with the EV Charger

Table 1 Inverter communication port definition

Inverter Type	Model	Pin Definition
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String Inverter	X3-MIC G2 X1-Mini G3 X1-Boost G3	Com Port Pin4&Pin5
	X1-Mini G4 X1-Boost G4	Com Port P3 Subport Pin7&Pin8
	X1-Smart G2	COM/CT Port Pin4&Pin5
	X3-Pro G2	RS485 Port Pin1&Pin2
Hybrid Inverter	X1-Hybrid G4 X3-Hybrid G4	COM Port Pin4&Pin5
	X1-IES X3-IES	COM Port P3 Subport Pin7&Pin8
	X3-HYB G4 Pro	COM2 Port EVC Subport Pin4&Pin5
	X1-Vast	COM Port EVC/RTU485 Subport Pin4&Pin5
	X3-Ultra	COM1 Port RS485 Subport Pin3/4&Pin5/6

4 EV Charger Settings

4.1 Home Edition

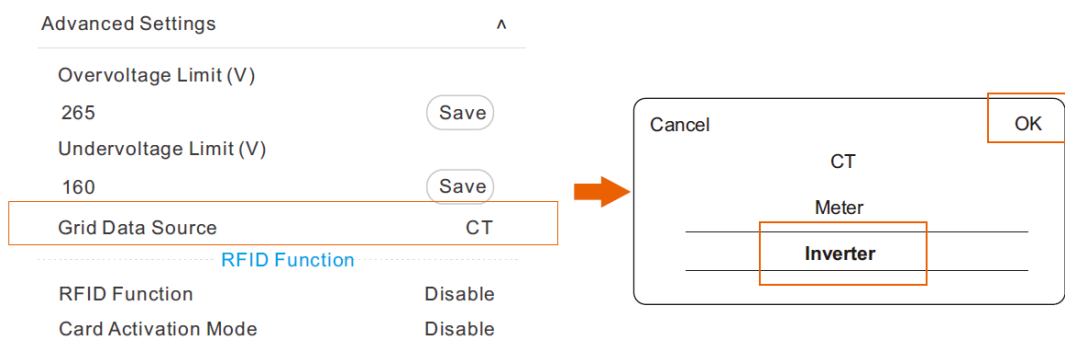


Figure 15 Home Edition Advanced Setting

- Select "Advanced Settings" and set the "Grid Data Source" as "Inverter" as shown above and touch "OK" to confirm. The buzzer of the EV-Charger will beep after the setting is succeeded.

4.2 Commercial Edition

Advanced Settings	
Application Scenarios	Private Charger
Start time	00:00
End time	00:00
Main Breaker Limit(A)	60
	<button>Save</button>

Grid Data Source

Cancel

OK

Private Charger

Commercial Charger(OCPP)

Figure 16 Commercial Edition Advanced Setting

- Select “Advanced Setting” and set the “Application Scenarios” as “Private Charger” as shown above and touch “OK” to confirm.

Grid Data Source	MODBUS
Slave Address	70
	<button>Save</button>
Timezone	(UTC) Coordinated Universal Time

Figure 17 Grid Data Source Setting

- Set the “Grid Data Source” as “MODBUS” as shown above. “Slave Address” only need to be set when multiple EV chargers are used in parallel.

4.3 Fusion Edition

< Advanced setting	
Application scene	Home
Alarm setting	160-265 v
Charging restrict	>
Random charging delay	Disable
Three phase imbalance	Disable
Restore factory settings	<button>Save</button>
EV Charger Reset	<button>Save</button>
Parallel Setting	Disable 0:0

Figure 18 Fusion Edition Advanced Setting

- Select “Advanced setting” and set “Application scene” as “Home” as shown above.

< Charger setting	
Activation mode	Plug&Charge
Work mode setting	Green:3A ECO:6A
Dynamic load balance	Disable
Modbus Setting	70/9600
LCD Language	English

Figure 19 Fusion Edition Charger Setting

- Select "Charger Setting" and set "Modbus Setting" according to the setting in the inverter.

5 Note

All the operations described in this guide can only be performed by qualified electricians with a good knowledge of the characteristics and maintenance of the smart EV-Charger and the inverter. Any operations are prohibited before reading this guide and the relevant manual carefully.

All the settings for the smart EV-Charger and the inverter need to refer to the corresponding manual strictly. SolaX Power will not be responsible for any damages or harms caused by improper setting.